Fourier Transform

Definition: **Forward transform:** $$F(\omega) = \mathcal{F}\big(f(t)\big) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt$$

Inverse transform: $$f(t) = \mathcal{F}^{-1}\big(F(\omega)\big) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}d\omega$$

Properties: (α, β = constant)

1) Linearity:	$\alpha f(t) + \beta g(t) \overset{\mathcal{F}}{\leftrightarrow} \alpha F(\omega) + \beta G(\omega)$
2) Scaling:	$f(\alpha t) \overset{\mathcal{F}}{\leftrightarrow} \frac{1}{\lvert\alpha\rvert} F\left(\frac{\omega}{\alpha}\right)$
3) Time shifting:	$f(t-\alpha) \overset{\mathcal{F}}{\leftrightarrow} e^{-j\omega\alpha}F(\omega)$
NOTE:	$f(\beta t-\alpha) \overset{\mathcal{F}}{\leftrightarrow} \frac{e^{-j\omega\frac{\alpha}{\beta}}}{\lvert\beta\rvert} F\left(\frac{\omega}{\beta}\right)$ or $f(\beta(t-\alpha)) \overset{\mathcal{F}}{\leftrightarrow} \frac{e^{-j\omega\alpha}}{\lvert\beta\rvert} F\left(\frac{\omega}{\beta}\right)$
4) Frequency shifting:	$e^{jt\alpha}f(t) \overset{\mathcal{F}}{\leftrightarrow} F(\omega-\alpha)$
5) Time derivative:	$\frac{df(t)}{dt} \overset{\mathcal{F}}{\leftrightarrow} j\omega F(\omega)$
6) Frequency derivative:	$-jtf(t) \overset{\mathcal{F}}{\leftrightarrow} \frac{dF(\omega)}{d\omega}$
7) Time integration:	$\int_{-\infty}^{t} f(\tau)d\tau = f * u(t) \overset{\mathcal{F}}{\leftrightarrow} \frac{F(\omega)}{j\omega} + \pi\delta(\omega)F(0)$
8) Duality/Symmetry:	$F(t) \overset{\mathcal{F}}{\leftrightarrow} 2\pi f(-\omega)$
9) Time convolution:	$f * g(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau \overset{\mathcal{F}}{\leftrightarrow} F(\omega)G(\omega)$
10) Frequency convolution:	$2\pi f(t)g(t) \overset{\mathcal{F}}{\leftrightarrow} F * G(\omega)$
11) Conjugate function	$f^*(t) \overset{\mathcal{F}}{\leftrightarrow} F^*(-\omega)$
12) Rayleigh's theorem	$\int_{-\infty}^{\infty} \lvert f(t)\rvert^2 d\tau \overset{\mathcal{F}}{\leftrightarrow} \frac{1}{2\pi}\int_{-\infty}^{\infty} \lvert F(\omega)\rvert^2 d\omega$
13) Correlation:	$R_{fg}(t) = \int_{-\infty}^{\infty} f(t+\tau)g^*(\tau)d\tau \overset{\mathcal{F}}{\leftrightarrow} F(\omega)G^*(\omega)$

Fourier transform of common functions:

f(t)	F(ω)
$\delta(t)$	1
1	$2\pi\delta(\omega)$
$\cos(\alpha t)$	$\pi\delta(\omega+\alpha)+\pi\delta(\omega-\alpha)$
$\sin(\alpha t)$	$j\pi\delta(\omega+\alpha)-j\pi\delta(\omega-\alpha)$
$\mathrm{sgn}(t)=\begin{cases}-1 & ;t<0\\ 0 & ;t=0\\ 1 & ;t>0\end{cases}$	$\frac{2}{j\omega}$
$u(t)=\begin{cases}0 & ;t<0\\ 0.5 & ;t=0\\ 1 & ;t>0\end{cases}$	$\frac{1}{j\omega}+\pi\delta(\omega)$
$\Pi(t)=\begin{cases}1 & ;\lvert t\rvert\le 0.5\\ 0 & ;\lvert t\rvert>0.5\end{cases}$	$\frac{\sin(0.5\omega)}{0.5\omega}$
$\Lambda(t)=\begin{cases}1-\lvert t\rvert & ;\lvert t\rvert\le 1\\ 0 & ;\lvert t\rvert>1\end{cases}$	$\frac{\sin^2(0.5\omega)}{(0.5\omega)^2}$
$e^{-\alpha\lvert t\rvert};\alpha>0$	$\frac{2\alpha}{\alpha^2+\omega^2}$
$e^{-\alpha t}u(t);\alpha>0$	$\frac{1}{\alpha+j\omega}$
$e^{-\pi t^2}$	$e^{-\frac{\omega^2}{4\pi}}$